

Completeness of Bang Bang Backward Shooting for Viability Kernel Approximation

1 Setup

System dynamics (double integrator):

$$\dot{x} = \begin{pmatrix} \dot{q} \\ \ddot{q} \end{pmatrix} = \begin{pmatrix} 0 & I \\ 0 & 0 \end{pmatrix} \begin{pmatrix} q \\ \dot{q} \end{pmatrix} + \begin{pmatrix} 0 \\ I \end{pmatrix} u$$

where $q \in \mathbb{R}^n$ is the joint position, $\dot{q} \in \mathbb{R}^n$ is the joint velocity, and $x = (q, \dot{q}) \in \mathbb{R}^{2n}$ is the full state. We write

$$A = \begin{pmatrix} 0 & I_n \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ I_n \end{pmatrix}.$$

Since $A^2 = 0$, the matrix exponential is $e^{At} = I + At$ and the exact discrete time solution under constant control u over $[0, h]$ is:

$$x(h) = \underbrace{(I + Ah)}_{A_h} x(0) + \underbrace{\left(hI + \frac{h^2}{2}A\right)B}_{B_h} u = A_h x(0) + B_h u.$$

Explicitly,

$$q(h) = q(0) + \dot{q}(0)h + \frac{1}{2}u h^2, \quad \dot{q}(h) = \dot{q}(0) + u h.$$

Input constraint:

$$u \in U = [-a_{\max}, a_{\max}]^n$$

(box constraint on acceleration).

State constraint: $X \subset \mathbb{R}^{2n}$ is a compact set encoding joint position limits, velocity limits, self collision avoidance, etc. We do **not** require X to be convex.

Seed set (equilibria):

$$S = \{x \in X \mid \exists u \in U : f(x, u) = 0\}.$$

For the double integrator, this reduces to

$$S = \{(q, 0) : q \in \text{proj}_q(X)\} \cap X.$$

Viability kernel (continuous time):

$$V = \{x_0 \in X : \exists u(\cdot) \text{ measurable, } u(t) \in U \forall t, \text{ s.t. } x(t) \in X \forall t \geq 0\}.$$

Constrained backward reachable set of S :

$$\text{BRS}(S) = \{x \in X : \exists u(\cdot) \text{ measurable, } \exists T \geq 0, x(T) \in S, x(t) \in X \forall t \in [0, T], u(t) \in U \forall t\}.$$

Shrunk constraint set: Define the distance to the boundary in the $\|\cdot\|_\infty$ norm by

$$\text{dist}_\infty(x, \partial X) = \inf_{y \in \partial X} \|x - y\|_\infty.$$

For $\eta > 0$, define

$$X_\eta = \{x \in X : \text{dist}_\infty(x, \partial X) \geq \eta\}.$$

Robustly viable set:

$$V^{\text{rob}} = \bigcup_{\eta > 0} \text{BRS}_\eta(S)$$

where

$$\text{BRS}_\eta(S) = \{x \in X_\eta : \exists u(\cdot) \text{ measurable, } \exists T \geq 0, x(T) \in S, x(t) \in X_\eta \forall t \in [0, T], u(t) \in U \forall t\}.$$

In words, V^{rob} is the set of states from which the system can reach an equilibrium while maintaining a strictly positive clearance from the constraint boundary throughout the entire trajectory.

Assumption 1 (Standard). The dynamics are given by the double integrator with box input constraint $U = [-a_{\max}, a_{\max}]^n$ and compact state constraint X .

Assumption 2. The viability kernel equals the constrained backward reachable set of the equilibria:

$$V = \text{BRS}(S).$$

That is, a state is viable if and only if it can be steered to an equilibrium while satisfying all constraints. This assumption holds for fully actuated manipulators with joint wise position and velocity limits, where any viable state has sufficient control authority to decelerate to a gravity compensated rest configuration. Even when this assumption is not satisfied (for example, for systems with tight self collision constraints coupling multiple joints), the algorithm computes a control invariant subset of V .

2 Algorithm

1. Fix step size $h > 0$. Initialize anchor set $A_0 = S \cap X$.
2. At iteration k : for each anchor $x_a \in A_k$, for each vertex control

$$u \in \text{vert}(U) = \{-a_{\max}, +a_{\max}\}^n,$$

compute the backward predecessor

$$x^- = A_h^{-1}(x_a - B_h u).$$

3. If $x^- \in X$ and the entire forward trajectory segment from x^- to x_a under constant control u remains in X , add x^- to the anchor set:

$$A_{k+1} = A_k \cup \{x^-\}.$$

4. If $x^- \notin X$ or the trajectory exits X , discard.

5. Repeat with $M = 1$ (every intermediate point becomes an anchor).

Let $W^h = \bigcup_{k=0}^{\infty} A_k$ be the set of all points produced by the algorithm with step size h , and $W = \bigcup_{h>0} W^h$.

Remark 1 (Intermediate feasibility check). For the double integrator, the trajectory under constant u over $[0, h]$ is

$$q(t) = q^- + \dot{q}^- t + \frac{1}{2} u t^2, \quad \dot{q}(t) = \dot{q}^- + u t.$$

The velocity is linear and the position is quadratic in t , so the intermediate check can be performed efficiently by evaluating extrema of each coordinate over $[0, h]$.

3 Theorem

Theorem 1. *Under Assumptions 1 and 2:*

$$V^{\text{rob}} \subseteq \overline{W} \subseteq V.$$

4 Proof

The proof proceeds through four lemmas.

4.1 Lemma 1: $W \subseteq V$ (Soundness)

Lemma 1. *Every point produced by the algorithm is viable.*

Proof. By induction on iteration k .

Base case. $A_0 = S \cap X$. For any $(q, 0) \in S \cap X$, the control $u(t) \equiv 0$ yields $x(t) = (q, 0) \in X$ for all $t \geq 0$. Hence $(q, 0) \in V$.

Inductive step. Suppose $A_k \subseteq V$. Take $x^- \in A_{k+1} \setminus A_k$. By construction, there exists $u \in \text{vert}(U)$ such that:

- (a) Applying constant control u for duration h maps x^- to some $x_a \in A_k$:

$$x_a = A_h x^- + B_h u.$$

- (b) The entire continuous trajectory $\xi(t)$ from x^- to x_a under constant control u satisfies $\xi(t) \in X$ for all $t \in [0, h]$.

Since $x_a \in V$ (by inductive hypothesis), there exists a control $u_a(\cdot)$ keeping the trajectory in X for all $t \geq 0$ starting from x_a . Define the control for x^- :

$$\tilde{u}(t) = \begin{cases} u, & t \in [0, h), \\ u_a(t - h), & t \geq h. \end{cases}$$

The resulting trajectory starts at x^- , stays in X during $[0, h]$ by condition (b), reaches x_a at time h , then follows the viable trajectory of x_a staying in X for all $t \geq h$. Hence $x^- \in V$. $\square$

4.2 Lemma 2: Discretization convergence

Lemma 2. *For the double integrator with compact X and box constraint U , for any $x_0 \in V^{\text{rob}}$ with margin $\eta > 0$ (that is, $x_0 \in \text{BRS}_\eta(S)$), there exists $h_0 > 0$ such that for all $h \leq h_0$: there is a finite sequence of controls $(u_0, u_1, \dots, u_{K-1}) \in U^K$ with each u_i constant over $[ih, (i+1)h)$, and $x_K \in S$, with the continuous trajectory satisfying $x(t) \in X_{\eta/2}$ for all $t \in [0, Kh]$.*

Proof. By assumption, there exists a measurable control $u(\cdot)$ and $T > 0$ with $x(T) \in S$ and $x(t) \in X_\eta$ for all $t \in [0, T]$.

Step 1 (Piecewise constant approximation). The trajectory $\{x(t) : t \in [0, T]\}$ is a compact subset of X_η . For the linear system $\dot{x} = Ax + Bu$, the Gronwall inequality (using the vector $\|\cdot\|_\infty$ norm and its induced matrix norm) gives: if $\tilde{u}(\cdot)$ is any other admissible control with trajectory $\tilde{x}(t)$ starting from x_0 , then

$$\|x(t) - \tilde{x}(t)\|_\infty \leq e^{\|A\|_\infty T} \|B\|_\infty \|u - \tilde{u}\|_{L^1([0, T])}.$$

Since piecewise constant functions with values in U are dense in $L^1([0, T]; U)$ (as U is compact and convex, the average of u over each subinterval lies in U ; see for example Brezis, *Functional Analysis*, Theorem 4.12), for any $\delta > 0$ and sufficiently small $h > 0$, there exists a piecewise constant $\tilde{u}$ with step size h such that

$$\sup_{t \in [0, T]} \|x(t) - \tilde{x}(t)\|_\infty < \delta.$$

Choosing $\delta \leq \eta/4$ ensures $\tilde{x}(t) \in X_{\eta/2}$ for all $t \in [0, T]$ and $\|\tilde{x}(T) - x(T)\|_\infty < \eta/4$.

Step 2 (Terminal correction). Write $\tilde{x}(T) = (\tilde{q}_T, \tilde{v}_T)$. Since $x(T) \in S$ has zero velocity and $\|\tilde{x}(T) - x(T)\|_\infty < \eta/4$ from Step 1, we have $\|\tilde{v}_T\|_\infty \leq \eta/4$. Define

$$m = \max\left(1, \left\lceil \frac{\|\tilde{v}_T\|_\infty}{a_{\max} h} \right\rceil\right), \quad u_{\text{cor}} = -\frac{\tilde{v}_T}{m h}.$$

Then $\|u_{\text{cor}}\|_\infty \leq a_{\max}$, so $u_{\text{cor}} \in U$. Applying u_{cor} for m steps of size h brings the velocity to zero exactly. The correction duration is $\tau = mh \leq \|\tilde{v}_T\|_\infty / a_{\max} + h$, and the trajectory deviation during correction satisfies

$$\sup_{t \in [0, \tau]} \|v(t) - \tilde{v}_T\|_\infty \leq \|\tilde{v}_T\|_\infty, \quad \sup_{t \in [0, \tau]} \|q(t) - \tilde{q}_T\|_\infty \leq \frac{1}{2} \|\tilde{v}_T\|_\infty \tau.$$

For any tolerance $\epsilon > 0$, choose $h \leq \epsilon/2$ and ensure $\|\tilde{v}_T\|_\infty \leq a_{\max} \epsilon/2$ (possible by Step 1). Then $\tau \leq \epsilon$, so both component deviations can be made arbitrarily small.

By choosing h sufficiently small in Step 1 (hence making $\|\tilde{v}_T\|_\infty$ and τ sufficiently small), we can ensure

$$\sup_{t \in [0, \tau]} \|(q(t) - \tilde{q}_T, v(t) - \tilde{v}_T)\|_\infty \leq \eta/4.$$

Since $\tilde{x}(T) \in X_{\eta/2}$, the triangle inequality for dist_∞ implies that the entire correction segment lies in $X_{\eta/4} \subset X$. Finally, the terminal state is $(q_{\text{final}}, 0) \in S$.

The concatenated control is piecewise constant with step size h , the trajectory stays in $X_{\eta/4} \subset X_{\eta/2}$, and the endpoint is in S . $\square$

Remark 2. This lemma serves the sole purpose of converting continuous time robust viability into a finite piecewise constant trajectory with positive margin. For general nonlinear systems, the convergence of discrete viability kernels to the continuous one is established in Saint Pierre (1994, *Approximation of the viability kernel*, *Appl. Math. Optim.* 29:187–209) and Aubin & Frankowska (*Set Valued Analysis*, Chapter 10). Our self contained proof exploits the linear structure.

4.3 Lemma 3: Vertex controls span the single step reachable set

Lemma 3. *For the double integrator with constant control over one step of size h , the reachable set from any x_0 using $u \in U$ equals the convex hull of the reachable set using vertex controls $u \in \text{vert}(U)$.*

Proof. The state at time h under constant u is $x(h) = A_h x_0 + B_h u$. The map $u \mapsto A_h x_0 + B_h u$ is affine. Since $U = [-a_{\max}, a_{\max}]^n$ is a convex polytope and B_h is a linear map:

$$R_1(x_0, h) = A_h x_0 + B_h \cdot U = \text{conv}\{A_h x_0 + B_h v : v \in \text{vert}(U)\}.$$

We used $B_h \cdot U = \text{conv}(B_h \cdot \text{vert}(U))$, which follows from $U = \text{conv}(\text{vert}(U))$ and the fact that linear maps commute with convex hull:

$$B_h \cdot U = B_h \cdot \text{conv}(\text{vert}(U)) = \text{conv}(B_h \cdot \text{vert}(U)).$$

The same holds for the backward single step reachable set, since $x^- = A_h^{-1}(x_a - B_h u)$ is also affine in u . $\square$

4.4 Lemma 4: Multi step vertex control approximation (Balanced dithering)

Lemma 4. *Consider a K step piecewise constant trajectory from x_0 with controls $(u_0, \dots, u_{K-1}) \in U^K$ and step size h , whose continuous trajectory stays in X with margin $\mu > 0$ from ∂X in $\|\cdot\|_\infty$. For any $\epsilon > 0$, there exists a step size $h' \leq h$ with $h/h' \in \mathbb{N}$ and a sequence of vertex controls in $\text{vert}(U)$ with step size h' , such that:*

1. *The resulting trajectory starts at x_0 and ends within ϵ of x_K in $\|\cdot\|_\infty$.*
2. *The continuous trajectory under each constant vertex control sub step remains entirely in X .*
3. *Backward chain compatibility: the trajectory can equivalently be read as a backward chain. If $x_K \in S$, the terminal point of the vertex control trajectory can be placed in S exactly.*

We use $\|\cdot\|_\infty$ throughout this lemma, so that $\|v\|_\infty = a_{\max}$ for all $v \in \text{vert}(U)$.

Proof. It suffices to prove the single macro step case $K = 1$; the multi step result follows by induction with Gronwall error propagation.

Single macro step ($K = 1$). We must approximate the effect of constant control $u_0 \in U$ over one step of size h , using only vertex controls with a finer step size $h' = h/N$ with $N \in \mathbb{N}$.

Balanced dithering construction. For each coordinate $i \in \{1, \dots, n\}$, define

$$\lambda_i = \frac{u_{0,i} + a_{\max}}{2a_{\max}} \in [0, 1].$$

At sub step $k \in \{1, \dots, N\}$, set

$$w_{k,i} = \begin{cases} +a_{\max}, & \text{if } \lfloor k\lambda_i \rfloor > \lfloor (k-1)\lambda_i \rfloor, \\ -a_{\max}, & \text{otherwise.} \end{cases}$$

Then $w_k = (w_{k,1}, \dots, w_{k,n}) \in \{-a_{\max}, +a_{\max}\}^n = \text{vert}(U)$ for every k .

Partial impulse error bound. Define $e_k = w_k - u_0$ (componentwise) and $\sigma_k = \sum_{j=1}^k e_j$. We claim:

$$|\sigma_{k,i}| \leq 2a_{\max} \quad \text{for all } k = 1, \dots, N \text{ and all } i = 1, \dots, n. \quad (\star)$$

Among the first k sub steps, exactly $\lfloor k\lambda_i \rfloor$ assign $+a_{\max}$ to coordinate i . So

$$\sum_{j=1}^k w_{j,i} = \lfloor k\lambda_i \rfloor a_{\max} + (k - \lfloor k\lambda_i \rfloor)(-a_{\max}) = (2\lfloor k\lambda_i \rfloor - k) a_{\max}.$$

The target cumulative sum is $k u_{0,i} = k(2\lambda_i - 1) a_{\max}$, hence

$$\sigma_{k,i} = \sum_{j=1}^k (w_{j,i} - u_{0,i}) = 2a_{\max}(\lfloor k\lambda_i \rfloor - k\lambda_i).$$

Since $|\lfloor k\lambda_i \rfloor - k\lambda_i| < 1$, we obtain $|\sigma_{k,i}| < 2a_{\max}$, which implies $(\star)$ and $\|\sigma_k\|_{\infty} \leq 2a_{\max}$.

Error analysis via closed form expressions. Since $A^2 = 0$, the N sub step composition has exact closed forms. Let $w_1, \dots, w_N$ be the dithered vertex controls. Then:

$$\dot{q}(h) = \dot{q}(0) + h' \sum_{k=1}^N w_k, \quad q(h) = q(0) + \dot{q}(0) h + h'^2 \sum_{k=1}^N \left(N - k + \frac{1}{2}\right) w_k.$$

The original single step trajectory gives $\dot{q}(h) = \dot{q}(0) + h u_0$ and $q(h) = q(0) + \dot{q}(0) h + \frac{h^2}{2} u_0$. Using $\sum_{k=1}^N (N - k + \frac{1}{2}) = \frac{N^2}{2}$ and $h'^2 N^2 = h^2$, the original trajectory can equivalently be written as N sub steps all using u_0 .

Velocity error.

$$\Delta_v = h' \sigma_N, \quad \|\Delta_v\|_{\infty} \leq 2a_{\max} h'. \quad (1)$$

Position error. Define $\Delta_q = h'^2 \sum_{k=1}^N (N - k + \frac{1}{2}) e_k$. Apply Abel summation with $a_k = N - k + \frac{1}{2}$ (decreasing, $a_k - a_{k+1} = 1$, $a_N = \frac{1}{2}$):

$$\sum_{k=1}^N a_k e_k = a_N \sigma_N + \sum_{k=1}^{N-1} (a_k - a_{k+1}) \sigma_k = \frac{1}{2} \sigma_N + \sum_{k=1}^{N-1} \sigma_k.$$

Taking $\|\cdot\|_{\infty}$ and applying $(\star)$:

$$\left\| \sum_{k=1}^N a_k e_k \right\|_{\infty} \leq \frac{1}{2} \|\sigma_N\|_{\infty} + \sum_{k=1}^{N-1} \|\sigma_k\|_{\infty} \leq \frac{1}{2} \cdot 2a_{\max} + (N-1) \cdot 2a_{\max} \leq 2Na_{\max}.$$

Therefore:

$$\|\Delta_q\|_{\infty} \leq 2Na_{\max} h'^2 = 2a_{\max} h h'. \quad (2)$$

Total single step error. From (1) and (2):

$$\|\Delta_q\|_{\infty} \leq 2a_{\max} h h' = \frac{2a_{\max} h^2}{N}, \quad \|\Delta_v\|_{\infty} \leq 2a_{\max} h' = \frac{2a_{\max} h}{N}. \quad (3)$$

Both are $O(h')$ and vanish as $N \rightarrow \infty$.

Intermediate feasibility. At any discrete time $t = kh'$ with $1 \leq k \leq N$, the same Abel summation argument (with upper limit k) gives:

$$\|\hat{q}(kh') - q_{\text{orig}}(kh')\|_{\infty} \leq 2a_{\max}hh', \quad \|\hat{v}(kh') - v_{\text{orig}}(kh')\|_{\infty} \leq 2a_{\max}h'.$$

Within each sub step $[kh', (k+1)h']$, the trajectory is a parabola under constant control, so the continuous time deviation adds at most $O(h'^2)$ to position and $O(h')$ to velocity, both dominated by the bounds above. Therefore, over $[0, h]$:

$$\sup_{t \in [0, h]} \|\hat{q}(t) - q_{\text{orig}}(t)\|_{\infty} \leq 3a_{\max}hh', \quad \sup_{t \in [0, h]} \|\hat{v}(t) - v_{\text{orig}}(t)\|_{\infty} \leq 3a_{\max}h'.$$

Assume the original continuous trajectory satisfies a uniform margin:

$$\text{dist}_{\infty}(x_{\text{orig}}(t), \partial X) \geq \mu, \quad \forall t \in [0, h],$$

where $x_{\text{orig}}(t) = (q_{\text{orig}}(t), v_{\text{orig}}(t))$ and $\hat{x}(t) = (\hat{q}(t), \hat{v}(t))$. We use that $x \mapsto \text{dist}_{\infty}(x, \partial X)$ is 1 Lipschitz with respect to $\|\cdot\|_{\infty}$:

$$\text{dist}_{\infty}(\hat{x}(t), \partial X) \geq \text{dist}_{\infty}(x_{\text{orig}}(t), \partial X) - \|\hat{x}(t) - x_{\text{orig}}(t)\|_{\infty}.$$

Hence, if $\sup_{t \in [0, h]} \|\hat{x}(t) - x_{\text{orig}}(t)\|_{\infty} < \mu$, then $\hat{x}(t) \in X$ for all $t \in [0, h]$.

Using $\|\hat{x} - x_{\text{orig}}\|_{\infty} = \max(\|\hat{q} - q_{\text{orig}}\|_{\infty}, \|\hat{v} - v_{\text{orig}}\|_{\infty})$ and the bounds above:

$$\sup_{t \in [0, h]} \|\hat{x}(t) - x_{\text{orig}}(t)\|_{\infty} \leq 3a_{\max}h' \max(h, 1) = \frac{3a_{\max}h \max(h, 1)}{N}.$$

Choosing

$$N > \frac{3a_{\max}h \max(h, 1)}{\mu}$$

ensures the vertex control trajectory stays in X over $[0, h]$.

Each sub step is a valid algorithm operation. Every sub step uses fixed step size $h' = h/N$, a single vertex control $w_k \in \text{vert}(U)$, and the forward trajectory segment remains in X . With $M = 1$, every intermediate point becomes an anchor in $W^{h'}$.

Multi step case (induction on K). Suppose the claim holds for $(K-1)$ step trajectories. For a K step trajectory with margin μ , apply the single macro step construction to the first step: choose N_1 large enough that the vertex control sub steps produce $\tilde{x}_1$ with $\|\tilde{x}_1 - x_1\|_{\infty} < \epsilon_1 < \mu/3$, with the trajectory staying in X .

From x_1 , the remaining $(K-1)$ step trajectory has margin μ . By the inductive hypothesis, it can be approximated by vertex control sub steps from x_1 with error $\epsilon_2 < \mu/3$, staying in X with margin $\mu - \epsilon_2$.

Starting the same vertex control sequence from $\tilde{x}_1$ instead of x_1 introduces an additional deviation. Since the double integrator is linear and the control sequence is state independent, the trajectory deviation satisfies the Gronwall bound:

$$\|\tilde{x}(t) - x(t)\|_{\infty} \leq e^{\|A\|_{\infty}(K-1)h} \|\tilde{x}_1 - x_1\|_{\infty} \leq e^T \epsilon_1,$$

where $T = Kh$ and $\|A\|_{\infty} = 1$. The total endpoint error is $\|\hat{x}_K - x_K\|_{\infty} \leq e^T \epsilon_1 + \epsilon_2$. The trajectory from $\tilde{x}_1$ stays in X provided $e^T \epsilon_1 + \epsilon_2 < \mu$. Choosing $\epsilon_1 < \mu e^{-T}/3$ and $\epsilon_2 < \mu/3$ ensures this.

To achieve any target $\epsilon > 0$, set $\mu' = \min(\mu, 3\epsilon/2)$ and apply the above.

Backward chain compatibility. The backward single step reachable set from y is

$$\{y^- : y^- = A_h^{-1}(y - B_h u), u \in U\}.$$

Since $u \mapsto y^-$ is affine, Lemma 3 applies: the backward reachable set is the convex hull of vertex control predecessors. The balanced dithering construction works identically for backward sub steps (the schedule depends only on u). Starting from $x_K \in S \subseteq A_0$, each macro step is subdivided into N backward vertex control sub steps, each adding a new anchor to $W^{h'}$.

If $x_K \in S$, the approximation endpoint $\hat{x}_K$ satisfies $\|\hat{x}_K - x_K\|_\infty < \epsilon$. To place the terminal point exactly in S , apply the same terminal correction as in Lemma 2 (append a short deceleration segment using $u_{\text{cor}} \in U$ with finitely many additional steps). This correction is itself a valid extension of the backward chain and produces a point in $S \cap A_0$. $\square$

4.5 Main Theorem

Theorem 2. Let $W = \bigcup_{h>0} W^h$ where W^h is the set of all points produced by the algorithm with step size h and vertex control sampling. Then:

$$V^{\text{rob}} \subseteq \overline{W} \subseteq V.$$

Proof. $\overline{W} \subseteq V$ (**Soundness**). By Lemma 1, $W^h \subseteq V$ for each $h > 0$, so $W \subseteq V$. Since V is closed (as the viability kernel of a compact set under continuous dynamics; see Aubin, *Viability Theory*, 1991, Theorem 3.3.2), $\overline{W} \subseteq V$.

$V^{\text{rob}} \subseteq \overline{W}$ (**Completeness**). Take any $x \in V^{\text{rob}}$ and any $\epsilon > 0$. We show $B_\epsilon(x) \cap W \neq \emptyset$.

Step 1. Since $x \in V^{\text{rob}}$, there exists $\eta > 0$ such that $x \in \text{BRS}_\eta(S)$.

Step 2. By Lemma 2, for sufficiently small $h > 0$, there is a K step piecewise constant control $(u_0, \dots, u_{K-1}) \in U^K$ with step size h such that the trajectory from x reaches S exactly, and the continuous trajectory stays in $X_{\eta/2}$ (margin $\mu = \eta/2$ from ∂X).

Step 3. By Lemma 4 applied to this K step trajectory with margin $\mu = \eta/2$, there exists $h' = h/N$ and a vertex control backward chain from S producing $\hat{x}_0 \in W^{h'}$ with $\|\hat{x}_0 - x\|_\infty < \epsilon$, with all intermediate trajectory segments remaining in X .

Step 4. Hence $B_\epsilon(x) \cap W^{h'} \neq \emptyset$, so $B_\epsilon(x) \cap W \neq \emptyset$. Since $\epsilon > 0$ was arbitrary, $x \in \overline{W}$. $\square$

$V^{\text{rob}} \subseteq \overline{W} \subseteq V$

4.6 Corollary

Corollary 1. If $V = \overline{V^{\text{rob}}}$ (which holds when X is convex, or more generally when X has no bottleneck structures where all viable trajectories to S must pass through regions of vanishing clearance from ∂X), then $V = \overline{W}$.

Proof. $V^{\text{rob}} \subseteq \overline{W}$ implies $\overline{V^{\text{rob}}} \subseteq \overline{W}$. By assumption $V = \overline{V^{\text{rob}}}$, so $V \subseteq \overline{W}$. Combined with $\overline{W} \subseteq V$, we obtain $V = \overline{W}$. $\square$

5 Summary of Conditions

Condition	Role in proof	Status
Double integrator dynamics $\ddot{q} = u$	Lemmas 2, 3, 4 (nilpotent A , exact discrete time formula)	Assumption 1
$U = [-a_{\max}, a_{\max}]^n$ convex polytope	Lemma 3 (vertex enumeration), Lemma 4 (coordinate wise dithering)	Assumption 1
X compact (not necessarily convex)	Lemma 1 (closure of V), Lemma 2 (compactness of trajectory)	Assumption 1
$V = \text{BRS}(S)$	Converts viability to backward reachability	Assumption 2
Intermediate feasibility check	Lemma 1 (soundness), Lemma 4 (margin propagation)	Algorithm design
Exhaustive vertex sampling at each anchor	Lemma 4 (coverage of all vertex sub step sequences)	In the limit of infinite iterations
$M = 1$ (every step anchoring)	Lemma 4 (every sub step point is an anchor)	Algorithm design
$h \rightarrow 0$	Lemma 2 (piecewise constant approximation), Lemma 4 ($O(h')$ error)	In the limit
$V = \overline{V^{\text{rob}}}$ (no bottlenecks)	Corollary ($V = \overline{W}$)	Not required for main theorem

6 On the Role of V^{rob}

The robustly viable set V^{rob} admits a clear physical interpretation: it is the set of states from which the robot can reach a safe rest configuration while maintaining a strictly positive safety margin from all constraint boundaries throughout the maneuver. The gap $V \setminus \overline{V^{\text{rob}}}$ consists of states that are technically viable but require threading the needle, meaning every trajectory to an equilibrium must pass through regions with vanishing clearance from the constraint boundary.

We argue that V^{rob} is the operationally appropriate safety set for three reasons.

1. Robustness to perturbations. Real robotic systems are subject to modeling errors, sensor noise, communication delays, and external disturbances. A state in $V \setminus V^{\text{rob}}$ requires zero margin maneuvers and offers no tolerance for any perturbation. V^{rob} , by construction, is an open set: every state in V^{rob} has a neighborhood also contained in V^{rob} , providing inherent robustness to small disturbances. This openness follows from a Gronwall argument: if $x_0 \in V^{\text{rob}}$ with margin η , then for sufficiently small $\delta > 0$, any $x'_0 \in B_\delta(x_0)$ has a trajectory staying in $X_{\eta/2}$, hence $x'_0 \in V^{\text{rob}}$.

2. Rarity of bottleneck structures. For typical robot manipulators, the constraint set X is defined by joint position limits (box constraints), velocity limits (box constraints), and self collision constraints (inequalities on minimum distances between link geometries). Self collision constraints remove thick regions from configuration space (since collision bodies have positive volume), leaving

corridors of positive width. True bottleneck structures, where the free space narrows to zero width, require degenerate geometric alignments (for example, two links that can only pass each other in a single configuration). Such degeneracies are atypical in practice, and for generic robot geometries, $V = \overline{V^{\text{rob}}}$.

3. Consistency with safe control objectives. In safety critical control frameworks such as viability preserving control, the controller must maintain the system within the safe set despite worst case disturbances. A safe set with zero clearance passages cannot provide any disturbance margin at those passages, making it unsuitable for robust safety guarantees. Using V^{rob} as the safe set ensures that at every point, there exists a strategy that tolerates bounded perturbations, which aligns with the practical requirements of safety critical robot control.

7 Remark on Practical Computation

The theorem establishes $V^{\text{rob}} \subseteq \overline{W} \subseteq V$ as a limiting result: the union W is taken over all step sizes $h > 0$, and exhaustive vertex sampling at every anchor is assumed. For any fixed $h > 0$, the algorithm produces a finite inner approximation $W^h \subseteq V$, whose accuracy depends on h . The approximation error is governed by:

1. **Temporal discretization:** the piecewise constant approximation error from Lemma 2, which is $O(h)$ for trajectories with positive interior margin.
2. **Vertex control discretization:** the per macro step dithering error from using vertex controls instead of arbitrary controls, which is $O(h') = O(h/N)$ per macro step (Lemma 4).

In practice, one runs the algorithm with a fixed h and obtains a finite point cloud inner approximating V .

8 Remark on Applicability to General Manipulators

The theoretical analysis assumes the double integrator model $\ddot{q} = u$ with box input constraints on acceleration. For manipulators with configuration dependent dynamics:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = \tau, \quad \tau \in [\tau_{\min}, \tau_{\max}],$$

the double integrator model applies when a feedback linearization or computed torque control layer maps the system to acceleration space:

$$u = M(q)^{-1}(\tau - C(q, \dot{q})\dot{q} - g(q)).$$

In this case, the effective input constraint on $u = \ddot{q}$ becomes configuration dependent:

$$u \in U(q, \dot{q}) = \{M(q)^{-1}(\tau - C(q, \dot{q})\dot{q} - g(q)) : \tau \in [\tau_{\min}, \tau_{\max}]\}.$$

The key properties required by our proof are preserved under this transformation: $U(q, \dot{q})$ remains a convex polytope (since $M(q)$ is invertible and the map $\tau \mapsto u$ is affine), and the dynamics are affine in u . Lemma 3 holds at each configuration, but the vertex set changes with $(q, \dot{q})$. For sufficiently small sub step sizes h' , the variation of $U(q, \dot{q})$ within one macro step is small by continuity of $M(q)$, and the balanced dithering construction in Lemma 4 applies with an additional $O(h')$ error from the varying vertex set. A complete extension to state dependent polytopic input constraints is left to future work.